

The Collaborative Ocular Melanoma Study (COMS) Randomized Trial of Pre-enucleation Radiation of Large Choroidal Melanoma III: Local Complications and Observations Following Enucleation COMS Report No. 11

COLLABORATIVE OCULAR MELANOMA STUDY GROUP*

- **PURPOSE:** To summarize local complications and observations following enucleation of eyes with large choroidal melanoma that were reported prospectively at scheduled examinations of patients enrolled in the Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation.
- **METHODS:** Of 1,003 patients with large choroidal melanoma who were assigned randomly at time of enrollment to enucleation alone or to pre-enucleation radiation, 994 were treated as assigned. Complications and observations were reported to the Coordinating Center on standard

forms completed at the time of enucleation surgery and during the immediate 24-hour postsurgery period, 1 to 6 weeks following surgery, 6 and 12 months after enucleation, and at examinations scheduled at 12-month intervals thereafter.

- **RESULTS:** The most common perioperative complication was pain that prolonged hospital stay, which was reported for six patients (1%) who had standard enucleation and eight patients (2%) who had pre-enucleation radiation. Patients treated with pre-enucleation radiation had somewhat more complications reported at the examination 1 to 6 weeks after surgery than did patients treated with enucleation alone, 36 (8%) and 21 (4%), respectively ($P = .03$, χ^2 test), but all complications were minor. During follow-up, fewer biopsy-confirmed tumor recurrences in the orbit were observed among patients treated with pre-enucleation radiation than with enucleation alone (0 vs 5, respectively; $P = .03$, Fisher exact test). Patients treated with pre-enucleation radiation also had a lower incidence of severe ptosis than did patients treated with enucleation alone ($P = .007$, log rank test). Among 307 patients examined 5 years after enucleation, the most frequent complication reported at the 5-year examination was poor prosthetic motility for 24 patients (16%) treated with enucleation alone and 30 patients (19%) treated with pre-enucleation radiation.

Accepted for publication May 14, 1998.

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• **CONCLUSIONS:** Complications were infrequent during the 5-year period following enucleation surgery. Five-year incidence rates and prevalence at the 5-year examination of most complications were similar in the two treatment arms. There was no indication that pre-enucleation radiation had resulted in more serious complications. (Am J Ophthalmol 1998;126:362–372. © 1998 by Elsevier Science Inc. All rights reserved.)

FOR MORE THAN A CENTURY, ENUCLEATION HAS been the accepted treatment for primary choroidal melanoma and remains so for melanoma of large size. During the past 20 years, adjunctive treatment with radiation has been proposed in an effort to improve survival of patients with large choroidal melanoma.^{1–11} In 1986, under the sponsorship of the National Eye Institute of the National Institutes of Health, the Collaborative Ocular Melanoma Study (COMS) Group initiated accrual of patients to a randomized multicenter clinical trial to compare, with respect to length of remaining life, pre-enucleation radiation to no radiation of eyes undergoing standard enucleation for large choroidal melanoma. Pre-enucleation radiation of the globe and orbit was selected because of the success of presurgical radiation in reducing metastases and improving other outcomes following surgery for other tumors.^{12–21} However, radiation has been reported to affect healing rates and scar formation,^{22,23} even when delivered in relatively low doses. Thus, there was concern that radiation would increase the frequency or severity of local complications and that differences in complication rates between enucleation alone and pre-enucleation radiation potentially would affect quality of life and patient satisfaction. Whether the anticipated higher risk of local complications would be offset by increased length of survival was unknown.

A number of complications related to enucleation, the implant, and the prosthesis have been reported. Operative complications reported include hemorrhage, rupture of the globe, extraocular tumor extension, transection of tumor that invades the optic nerve, dehiscence in the orbital roof, and loss of a muscle during surgery.^{24,25} Early postoperative complications that have been reported include edema and ecchymosis of orbital tissue, acute orbital

infection, extrusion of the implant, allergic reaction to the implant material, and ptosis.^{24,25} Late postoperative complications have included discharge from the socket due to allergy, irritation, or infection; superior sulcus defect; ptosis of the upper eyelid, lower lid, or both; contracted socket; inversion of eyelid; enophthalmos; unacceptable or bizarre motility of the implant and prosthesis; implant displacement or migration (sometimes leading to loss of the implant); contracted fornices; granulation tissue; orbital cellulitis; and exposure or extrusion of the implant.^{24–29} However, few data have been published regarding the frequency of occurrence or incidence rates of these complications. Vistnes²⁷ and Vistnes and associates²⁸ reported that ptosis of the upper lid occurred in 12 (18%) of 66 cases and of the lower lid in 50% of anophthalmic orbits. Reports published during the past 50 years have focused largely on comparisons of types of implants^{26,29} or problems associated with one or more specific types of implants.^{30–35} Kreissig and associates³⁶ reported complications after preoperative external beam radiation in doses from 12 to 40 Gy among 15 of 26 patients with choroidal melanoma who returned for examination after enucleation. All 15 patients had enophthalmos; in 11, enophthalmos was greater than 2 mm. Four patients had reduced tear production in the anophthalmic socket in comparison to the fellow eye. No patients had loss of eyelashes or eyebrows. There was no local tumor recurrence; however, time from enucleation to examination was not given. Fan and Robertson³⁵ emphasized the need for long-term follow-up when evaluating orbital implants.

Enrollment in the COMS clinical trial of pre-enucleation radiation of large choroidal melanoma was completed in December 1994, when the target number of patients had been accrued. The initial mortality findings from the trial are reported elsewhere.³⁷ The purpose of this article is to compare complications and observations of the orbit and adnexa of the affected eyes during the perioperative and postoperative periods following enucleation in the two treatment arms. Our analysis is unique because all patients had a single diagnosis, that is, choroidal melanoma, data were collected prospectively, multiple surgeons performed the enucleations, radiation dose and treatment techniques

were standardized, and patients were assigned randomly to enucleation alone or to pre-enucleation radiation at time of enrollment, with stratification by clinical center, so that reports of observed effects of radiation on complications are not biased by patient selection, differences in length of follow-up, or differences in clinical criteria between treatment arms.

PATIENTS AND METHODS

MANY ASPECTS OF THE COMS DESIGN AND METHODS have been published.³⁷⁻⁴³ The complete *COMS Manual of Procedures*⁴⁴ and the *COMS Forms Book*⁴⁵ are available.

• **PATIENT ELIGIBILITY AND ENROLLMENT.** The eligibility and exclusion criteria, baseline examinations, and the procedures for patient enrollment and randomization are described elsewhere.^{37,38} The initial ophthalmic evaluation included external examination of the eyes and eyelids, ophthalmoscopic examination of the tumor and of the fundus of each eye, and standardized A-scan and B-scan echography. Patients in whose eyes extrascleral extension of 2 mm or more was visible on echography or clinical examination prior to enrollment were ineligible. The clinical center ophthalmologist was notified whenever extrascleral extension was suspected at the COMS Echography Center based on review of photoechograms. Patients who were judged by the COMS ophthalmologist to be eligible after completion of all baseline evaluations were invited to participate in this clinical trial. Patients who gave signed consent were assigned randomly to enucleation alone or to pre-enucleation radiation by personnel at the COMS Coordinating Center.

• **PRE-ENUCLEATION RADIATION.** Radiotherapy was initiated as soon as possible after randomization. The radiotherapy protocol specified that the prescribed dose of 20 Gy be delivered in five daily fractions of 4 Gy each, that radiotherapy be completed within a maximum of 8 days, and that the eye be enucleated on the day the last fraction was delivered whenever possible but no more than 80 hours later.⁴⁴

Photon beams from either cobalt-60 or accelerators of 4 to 10 MV were acceptable for administering radiation. Multiple fields with wedges were recommended; a single appositional field was acceptable. When wedged pairs were used, the dose was prescribed at the intersection of the axes of the two fields. Whenever a single appositional field was used, the dose was prescribed at 2-cm depth on the central axis of the beam. The target volume included the choroidal melanoma, the posterior half of the globe, and a 1-cm margin lateral and posterior to the globe. Bolus was used for 10-MV photons and in situations in which the dose prescription point otherwise would be in the dose buildup region.

• **ENUCLEATION SURGERY.** The COMS protocol specified that enucleation of the tumor-containing eye take place within 4 weeks after randomization, and recommended enucleation within 2 weeks whenever possible. Enucleation surgery was permitted under either general or local anesthesia.

As specified by the COMS protocol,⁴⁴ after a 360-degree limbal peritomy and detachment of the conjunctiva and Tenon's capsule in each quadrant under direct visualization, the surgeon inspected the quadrants for evidence of extraocular extension. All visible tumor was dissected without disturbance. Extraocular muscles were cut from insertions with as little manipulation as possible; the optic nerve was cut with a scissors beyond any visible tumor extension along the nerve as far posteriorly in the orbit as appropriate without performing a lateral orbitectomy. The protocol specified that, after removing the eye, the surgeon should close rents in Tenon's capsule and insert a nonmetallic implant; the opening in posterior Tenon's capsule was not closed routinely. A tenonectomy was performed whenever the surgeon observed tumor in the sclera or tumor extension beyond the scleral surface.

Ocular and systemic complications observed by the COMS surgeon during surgery or upon examination in the 24-hour period immediately following surgery were reported. Information was not sought from the anesthetist or from others who may have provided care to the patient.

• **CENTRAL HISTOPATHOLOGIC REVIEW.** Enucleated eyes were sent to the COMS Pathology Center

for processing or were processed locally, depending upon the availability of a COMS pathologist.

The details of processing, slide preparation, and central review have been described elsewhere.^{40,41} Sets of microscopic slides were sent to the members of the COMS Pathology Review Committee for confirmation of the diagnosis and for recording descriptive features of the choroidal melanoma. Of relevance to this report, each member of the Pathology Review Committee assessed from the slides the degree of scleral invasion of the tumor and classified invasion as none, within the sclera but not to the scleral surface, to the scleral surface but not extrascleral, extrascleral and wholly within the specimen, or extrascleral and presumably transected during surgery.

- **PATIENT FOLLOW-UP.** Patients were scheduled for a posttreatment examination 1 to 6 weeks after surgery. Clinical examinations for data collection purposes were scheduled 6 and 12 months after enrollment and annually thereafter; examinations at the midpoint between annual examinations were optional. At each COMS examination, the socket and eyelids were scrutinized for local tumor recurrence and radiation treatment effects. Tumor recurrence in the orbit was suspected whenever there was pigmented tissue or a visible or palpable nodule in the orbit. The fit of the prosthesis was checked and the patient was referred to the ocularist for modifications as necessary. A finding of severe ptosis was based on the clinical judgment of each examining ophthalmologist and not on intrapalpebral fissure measurements or levator function.

- **DATA MANAGEMENT AND ANALYSIS.** Findings from each examination were recorded on standard forms⁴⁵ and forwarded to the COMS Coordinating Center; this analysis is based on data received by July 31, 1997. Fisher exact tests and χ^2 tests⁴⁶ were used to compare frequencies of complications between treatment arms. The product-limit method⁴⁷ was used to estimate time to first report of individual complications; treatment groups have been compared using the log rank test.^{48,49} Potential risk factors for specific complications were evaluated by comparing time to first report of individual complications within subgroups of patients. Cox propor-

tional hazards regression models,⁵⁰ using SAS (SAS, Inc, Cary, North Carolina), were employed to adjust simultaneously for several baseline and treatment characteristics, to identify any that were important predictors of severe ptosis, and to estimate the hazard ratio (risk ratio) and 95% confidence interval (CI) for severe ptosis following pre-enucleation radiation compared to enucleation alone. Other complications were too infrequent to permit multivariate analyses.

RESULTS

OF THE 1,003 PATIENTS ENROLLED IN THIS COMS CLINICAL trial, 506 were assigned to standard enucleation and 497 to pre-enucleation radiation. The two treatment arms were well balanced with respect to sociodemographic characteristics, medical and ocular history, and tumor characteristics.³⁷ Mean age of these patients was 60 years; 57% were male; 97% were non-Hispanic whites. Mean apical height was 9.5 mm; mean longest basal diameter was 17.3 mm. A history of one or more nonmalignant chronic conditions was reported by 67% of patients at the time of enrollment.³⁷

- **ADHERENCE TO TREATMENT PROTOCOLS.** Of the 497 patients assigned to pre-enucleation radiation, 491 were treated as assigned. Of the 506 patients assigned at random to standard enucleation, 503 were treated as assigned. These 994 patients are the subjects of this report.

Adherence to the radiotherapy protocol was excellent.³⁷ Five of the 491 eyes treated with pre-enucleation radiation received slightly less than 20 Gy; 43 received somewhat more than 20 Gy. The dose delivered was less than 18 Gy or greater than 22 Gy for only six eyes. The total dose was delivered in five fractions in all cases. Radiation was delivered over 5 consecutive days for 193 (39%) of patients and over more than an 8-day period for only three patients. Only nine of the 491 radiation treatments (2%) were judged centrally to be a major deviation from the protocol.³⁷

More than three quarters of all patients (763 [77%]) had enucleation during the first 2 weeks after enrollment and 973 (98%) during the first 4 weeks,

TABLE 1. Patients Who Had Tenonectomy by Histopathologic Finding Regarding Scleral Invasion

Degree of Scleral Extension, Based on Histopathology	No. of Patients	Tenonectomy Performed (No. [%])
None	345	3 (1)
Within sclera	568	14 (2)
Extrascleral	80	17 (21)
Total	993*	34 (3)

*One eye was not available for histopathologic evaluation.

as recommended and required, respectively, by the COMS protocol. Only 21 eyes, four treated with enucleation alone and 17 treated with pre-enucleation radiation, were removed more than 4 weeks after enrollment.

Enucleations were performed by 110 different surgeons. The number of enucleations per surgeon ranged from one to 42, with a median of five. Twenty surgeons who each performed 18 or more enucleations accounted for 515 enucleations (52%). Only two enucleations were performed by surgeons who had not yet met COMS certification requirements.^{39,44}

• **PERIOPERATIVE AND IMMEDIATE POSTOPERATIVE COMPLICATIONS.** Extension of the choroidal melanoma of more than 2 mm beyond the sclera that had not been suspected at the time of patient enrollment was observed during enucleation of 23 eyes, nine (2%) that were treated with standard enucleation (with the extension encapsulated in seven eyes) and 14 (3%) treated with pre-enucleation radiation (with the extension encapsulated in eight eyes). A tenonectomy was performed in 16 eyes (3%) treated with standard enucleation and in 18 eyes (4%) given pre-enucleation radiation because the surgeon observed or suspected extrascleral extension of the tumor of any size. The surgeon's decision to perform a tenonectomy was supported by review of histopathology in all but three instances (Table 1).

Complications reported by the surgeon during surgery or at the time of examination within 24 hours after surgery are summarized in Table 2. Only 36 patients (4%) had any surgical or anesthetic complication reported by the surgeon.

TABLE 2. Patients With Complications Noted Intraoperatively or Within the 24-Hour Postoperative Period, by Treatment Arm

Perioperative Complication	Standard Enucleation (N = 503) (No. [%])	Pre-enucleation Radiation (N = 491) (No. [%])
Anesthetic complication	1 (<1)	0
Surgical complications		
Pain that prolonged hospitalization	6 (1)	8 (2)
Hemorrhage	6 (1)	4 (1)
Nausea or vomiting	3 (1)	1 (<1)
Cardiovascular or pulmonary problems	1 (<1)	4 (1)
Urinary retention	0	1 (<1)
Fever or elevated temperature	1 (<1)	1 (<1)
Local surgical problem	0	1 (<1)
Any surgical complication*	16 (3)	19 (4)
Total, any anesthetic or surgical complication*	17 (3)	19 (4)

*Some patients had more than one complication reported.

Of 964 patients examined during 1 to 6 weeks after enucleation, one or more complications were reported for 21 patients (4%) treated with enucleation alone and for 36 patients (8%) treated with pre-enucleation radiation (Table 3; $P = .03$, χ^2 test). The most frequent reports were of pain requiring prolongation of hospitalization (nine patients in each treatment arm). Other complications were reported for five or fewer patients total. Eyelid swelling was reported for three patients assigned to pre-enucleation radiation but for no patient assigned to enucleation alone ($P = .08$, Fisher exact test). Implant displacement was reported for two patients treated with standard enucleation but for none of those treated with pre-enucleation radiation. Although conjunctival wound dehiscence was reported for four patients treated with pre-enucleation radiation and only one treated with enucleation alone, the difference was not statistically significant ($P = .17$, Fisher exact test). Other problems voluntarily reported by the surgeon (Table 3) were somewhat more frequent among patients assigned to pre-enucleation radiation ($P = .07$, Fisher exact test); conformer problems were those most often reported in this category.

• **FOLLOW-UP FOR COMPLICATIONS.** Follow-up ophthalmologic examination rates were similar in

TABLE 3. Patients With Complications Reported 1 to 6 Weeks After Enucleation

Complication	Standard Enucleation (N = 490) (No. [%])	Pre-enucleation Radiation (N = 474) (No. [%])
Pain requiring medication	9 (2)	9 (2)
Conjunctival wound dehiscence	1 (<1)	4 (1)
Infection	1 (<1)	3 (1)
Decreased facial sensation	2 (<1)	1 (<1)
Eyelid swelling	0	3 (1)
Inflammation	1 (<1)	1 (<1)
Implant displacement	2 (<1)	0
Loss of scalp hair	0	2 (<1)
Ptosis	0	2 (<1)
Conjunctival chemosis	1 (<1)	1 (<1)
Ecchymosis	0	2 (<1)
Systemic complications	1 (<1)	1 (<1)
Loss of eyebrow or eyelashes	0	1 (<1)
Orbital or conjunctival hemorrhage	0	1 (<1)
Other hemorrhage	1 (<1)	0
Other	3 (1)	9 (2)
Conformer problems	1 (<1)	4 (1)
Local skin allergy	0	2 (<1)
Other eyelid problems	2 (<1)	0
Tissue edema	0	1 (<1)
Friable tissue	0	1 (<1)
Tearing	0	1 (<1)
Total, any complication*	21 (4)	36 (8)

*Some patients had more than one complication reported.

the two treatment arms (Table 4). About 40% of all patients treated initially and nearly three quarters of patients still alive had completed the 4-year examination; nearly one third of all patients treated and 70% of all still living had completed the 5-year examination.

• **LOCAL TUMOR RECURRENCE.** Observations and complications reported by the examining ophthalmologist at least once from the 6-month examination through the 5-year examination of each patient are summarized in the first two columns of Table 5. Definite or suspected tumor recurrence in the orbit was reported for six patients (1%) treated with enucleation alone but only one patient treated with pre-enucleation radiation. Biopsy findings from these seven cases confirmed local recurrence in five of six patients treated with enucleation alone but not in the patient treated with pre-enucleation

TABLE 4. Scheduled Follow-up Examinations Expected and Completed During 5 Years After Enrollment

Time Since Enrollment (mos)	Standard Enucleation (n = 503)		Pre-enucleation Radiation (n = 491)	
	Expected* (No.)	Completed (No. [%])	Expected* (No.)	Completed (No. [%])
6	498	469 (94)	482	456 (95)
12	483	447 (93)	467	430 (92)
24	427	371 (87)	417	363 (87)
36	368	299 (81)	364	280 (77)
48	281	205 (73)	291	213 (73)
60	209	147 (70)	225	161 (72)

*Number of surviving patients eligible for the specified examination by the analysis date based on time since enrollment.

radiation ($P = .03$, Fisher exact test). None of the six cases treated with enucleation alone was noted by the surgeon to have extrascleral extension; histopathologic review of these cases showed tumor extension to the scleral surface in one eye but without suspicion of residual orbital tumor. The five patients in whom orbital recurrence was confirmed by biopsy had metastatic melanoma diagnosed before the orbital recurrence was reported and died less than 1 year after local recurrence. The two patients in whom orbital recurrence was not confirmed by biopsy died within 2 years of the report of suspected recurrence, with metastatic melanoma diagnosed in each case approximately 1 year after suspected tumor recurrence was observed in the orbit.

• **OTHER COMPLICATIONS DURING FOLLOW-UP.** The observation reported for the greatest number of patients was poor motility of the prosthesis, for 124 patients (26%) treated with enucleation alone and 133 patients (29%) treated with pre-enucleation radiation (Table 5). The next most common observation was poor alignment of the prosthesis, reported at least once for 88 patients (18%) and 81 patients (18%), respectively, in the enucleation alone and pre-enucleation treatment arms. Loss of eyelashes (Table 5) was the only complication reported more frequently among patients in the pre-enucleation radiation arm to a statistically significant degree ($P = .03$, χ^2 test). Implant displace-

TABLE 5. Patients With Postenucleation Complications Ever Reported by 5 Years After Surgery and Those Reported at 5-Year Examination

Postenucleation Complication	Ever Reported, 6-Month Through 5-Year Examinations (No. [%])		Reported at 5-Year Examination (No. [%])	
	Standard Enucleation (N = 487)	Pre-enucleation Radiation (N = 474)	Standard Enucleation (N = 147)	Pre-enucleation Radiation (N = 160)
Poor prosthetic motility	124 (26)	136 (29)	24 (16)*	30 (19)*
Poor alignment of prosthesis	88 (17)	81 (17)	7 (5)*	3 (2)*
Severe ptosis	49 (10)	24 (5)	6 (4)	3 (2)
Displacement of implant	15 (3)	15 (3)	0	1 (1)
Loss of eyelashes (moderate or severe)	8 (2)	19 (4)	1 (1)	1 (1)
Eyelid ulceration	6 (1)	9 (2)	0	1 (1)
Tumor recurrence in orbit				
Suspected	6 (1)	1 (<1)	0	0
Confirmed by biopsy	5 (1)	0	0	0

*One patient not wearing prosthesis excluded from denominator.

ment and eyelid ulceration were reported with similar frequency in the two treatment arms during the first 5 years after enucleation.

Severe ptosis was reported at least once during follow-up for twice as many patients treated with enucleation alone as for patients treated with pre-enucleation radiation (Table 5; $P = .003$, χ^2 test). This difference has persisted throughout the follow-up period (Figure; $P = .001$, log rank test). Adjusting only for time since enrollment, the risk ratio estimate for severe ptosis among patients treated with pre-enucleation radiation vs those treated with enucleation alone was 0.54 (95% CI, 0.34 to 0.85). Reporting patterns were examined in detail in an effort to identify any systematic bias by examining ophthalmologist, clinical center, or other factors. However, no evidence of bias in reporting this complication was uncovered. Severe ptosis was reported only at the 6-month examination for seven patients treated with standard enucleation and six patients treated with pre-enucleation radiation. For 36 additional patients, 25 of whom underwent standard enucleation and 11 who had pre-enucleation radiation, severe ptosis was reported at a single examination at 1 year or longer after surgery. Severe ptosis was reported at two or more examinations for only 24 patients, 17 in the standard enucleation arm and seven in the pre-enucleation radiation arm.

A number of potential explanatory variables were examined in an attempt to elucidate the difference in the incidence of severe ptosis between the two treatment arms. Age and gender of the patient, smoking history, tumor size, time since diagnosis of choroidal melanoma, and tenonectomy at time of enucleation were included as covariates in a Cox proportional hazards model, together with treatment arm, for evaluation of their effects. Only treatment arm and tenonectomy had statistically significant influences on the incidence of severe ptosis after adjustment for all covariates listed above. Pre-enucleation radiation reduced the incidence of severe ptosis ($P = .005$; estimated risk ratio = 0.52, 95% CI, 0.33 to 0.82). Tenonectomy increased the incidence of severe ptosis ($P = .006$; estimated risk ratio = 3.1, 95% CI, 1.4 to 6.9), despite the fact that relatively few patients (34 total [3%]) had a tenonectomy. Source of radiation and method of radiation delivery also were examined together with other covariates within the subgroup of patients treated with pre-enucleation radiation but were found to have no influence on the incidence of severe ptosis in this treatment arm.

As shown in the last two columns of Table 5, the prevalence rates of most complications at the 5-year examination were quite low. The most frequently reported problems among patients who had enrolled early enough and survived long enough to be eligi-

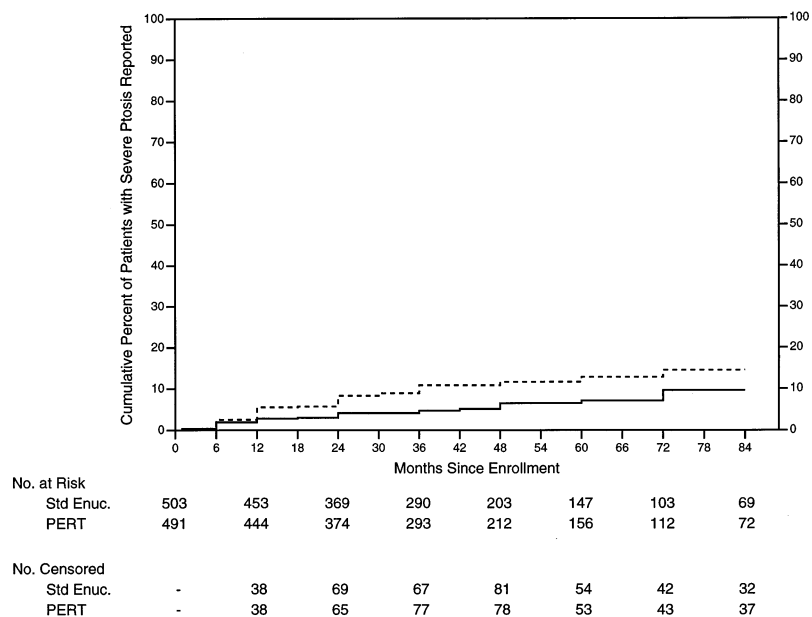


FIGURE. Cumulative percentage of enucleated eyes with severe ptosis reported by time since enrollment. Solid line = eyes treated with pre-enucleation radiation (n = 491); dashed line = eyes treated with enucleation alone (n = 503). Numbers of eyes at risk at selected times after enrollment and numbers censored due to deaths and time since enrollment given by treatment arm below horizontal axis.

ble for the 5-year examination were motility and alignment of prostheses; prevalence rates were similar in the two treatment arms.

Although the estimated cumulative 5-year incidence of severe ptosis was 13% and 7% in patients treated with standard enucleation and pre-enucleation radiation, respectively (Figure), the observed prevalence of ptosis 5 years after surgery was 4% and 2%, respectively, as shown in the last two columns of Table 5. Information was not collected regarding corrective surgery. Thus, it is unknown how often ptosis resolved spontaneously and how often surgical correction was performed.

DISCUSSION

WHEN THE COMS TRIAL OF PRE-ENUCLEATION RADIATION of large choroidal melanoma was undertaken, there was some concern that the radiation would cause local problems because of damage to the orbital tissue, conjunctiva, and eyelids. Data collection regarding possible complications focused on those that were believed to be the most serious at the time the trial was designed, that is, local tumor

recurrence, loss of the implant, displacement of the implant, eyelid ulceration, and selected cosmetic problems, including loss of eyelashes, poor motility and alignment of the prosthesis, and severe ptosis. No information was available in the published literature at that time regarding complications following pre-enucleation radiation. In an earlier article,⁴³ we reported preliminary findings regarding complications of enucleation observed during surgery or at the postenucleation examination 1 to 6 weeks after surgery. However, the very early complications were not reported by treatment arm because accrual to the COMS clinical trial of pre-enucleation radiation had not been completed. Based on the data that have been collected and presented in this report, with 3 or more years elapsed since time of enrollment for almost all patients, no harmful effects of pre-enucleation radiation were observed. The COMS trial has demonstrated that there were somewhat more very early complications following pre-enucleation radiation than following enucleation alone, but these were all minor, for example, eyelid swelling (Table 3). Tumor recurrence in the orbit was more frequent in patients treated with enucleation alone (Table 5).

Although information was not collected about corrective surgery, replacement of implants, or replacement or revision of prostheses, there is no indication that these procedures were necessary more often in the patients treated with pre-enucleation radiation than among patients treated with enucleation alone.

The data on which this report is based were obtained from nearly 6,300 examinations by COMS ophthalmologists and surgeons. Because no attempt was made to collect information concerning complications from other patient caregivers, the actual incidence rates may be underestimated, particularly for systemic complications associated with the initial surgical procedure. Nevertheless, underreporting should apply equally to both treatment arms. Because patients were assigned randomly to treatment within each clinical center, it is unlikely that the few differences in complication rates observed between treatment arms are attributable to different examiners or examiner biases. Furthermore, examination rates were similar in the two treatment groups.

The importance of scheduled prospective follow-up to document complications among patients undergoing enucleation is underscored by the differences in 5-year incidence rates of complications vs 5-year prevalence rates (Table 5). Only patients who enrolled at least 5 years before the analysis cutoff date, who survived for 5 years, and who were able to return for a 5-year examination contributed to the 5-year prevalence data. With 60% of patients in this clinical trial alive 5 years after initial treatment,³⁷ the low rate of complications observed at the 5-year examination are reliable estimates and provide reassurance to patients who undergo enucleation for choroidal melanoma that early local complications, apart from orbital recurrence, are transient or correctable.

Hydroxyapatite implants were introduced during the course of this COMS clinical trial. As reported by Hornblass and associates,²⁹ hydroxyapatite orbital implants have gained in popularity since their approval by the Food and Drug Administration in 1989. Most recent reports of enucleation-associated complications have focused on hydroxyapatite implants.^{30-34,51} Information regarding type of implant was reported for only the last 198 patients (20%) enrolled in the COMS

clinical trial of pre-enucleation radiation. Of these patients, 99 (50%) received hydroxyapatite implants, 50 (25%) received methylmethacrylic implants, 37 (19%) received silicone implants, and 12 (6%) received other types. Thus, too few COMS patients were known to have had hydroxyapatite implants, and those patients have been followed for too short a time to permit reliable comparisons of complication rates by implant at this time. With longer follow-up of surviving COMS patients, some comparisons by implant type may be possible within the subgroup of 198 patients who enrolled during 1993 and 1994, when information regarding implant type was collected.

Initial mortality findings published in an earlier article³⁷ provide no evidence of a difference between pre-enucleation radiation and enucleation alone with respect to patient survival through 8 years of follow-up. Our analysis documents that pre-enucleation radiation does not increase the risk or severity of most complications during enucleation surgery or during 5-year follow-up. Orbital tumor recurrence was reported and confirmed for fewer patients treated with pre-enucleation radiation than with enucleation alone. Severe ptosis also was reported for fewer patients treated with pre-enucleation radiation. To the best of our knowledge, this observation has not been reported by others. Other comparative investigations of pre-enucleation radiation have been too small to have detected a difference between treatment arms. Although tenectomy was a statistically significant predictor of later ptosis, there was no difference in tenectomy rates between treatment arms and no evidence from statistical modeling of an interaction between tenectomy and treatment arm. A possible explanation of the ptosis finding is that minimal fibrosis of the levator complex was induced in some eyelids by pre-enucleation radiation. Another possible explanation is that patients who were treated with pre-enucleation radiation suffered less damage to third cranial nerve branches during surgery. However, because reports of severe ptosis were based on clinical assessment and not on standardized methods, it is difficult to know how much weight to give to this finding of decreased risk of severe ptosis in patients treated with pre-enucleation radiation.

The COMS experience provides the most unbiased comparison of complications following stan-

dard enucleation and pre-enucleation radiation in the largest group of patients reported who had primary enucleation for choroidal melanoma. In addition, prospective follow-up of almost all living patients for 5 years or longer has provided the most complete, systematically collected data reported to date. In the absence of a documented survival benefit from pre-enucleation radiation, selection of primary treatment of large choroidal melanoma in future patients is likely to be based on consideration of cost, time required for treatment and posttreatment care, likelihood of further local treatment as a consequence of complications, and patient preference.⁵² Continued prospective follow-up examinations of patients in this COMS clinical trial, expected for a minimum of 8 years for the last patients enrolled, will provide documentation of the long-term risks of complications among survivors in the two treatment arms so that future patients and their physicians can make informed decisions about treatment of large choroidal melanoma.

ACKNOWLEDGMENT

The COMS investigators are grateful to the ophthalmologic community for supporting COMS clinical trials of radiotherapy by referring patients with choroidal melanoma to participating clinical centers. The investigators are particularly grateful to the patients and their families who have participated for the benefit of future patients.

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